

Appendix C

III-V (GaAs)*

Silicon has historically dominated integrated circuit technology development as the primary semiconductor material.

The principal focus in recent years on a silicon alternative has concentrated on valence III-V compounds, such as gallium arsenide (GaAs), as a substrate material. As a semiconductor material, GaAs exhibits increased capabilities over silicon, such as electron mobility five to six times that of silicon. This characteristic, coupled with the semi-insulating substrate of GaAs, leads to increased performance in both speed and power consumption.

Gallium arsenide has a zinc blende structure consisting of two interpenetrating face centered cubic (*fcc*) sublattices which relate to the growth of high quality ingot material. The technology involved in the growth of GaAs is considerably more complicated than that employed for silicon, as one is dealing with a more complicated two-phase equilibria and a highly volatile component, arsenic (As). The precise control of the As vapor pressure in the ingot growth system is required to maintain exact stoichiometry of the GaAs compound during the growth process.

* This information is compiled primarily from references F1, F2, and F3 shown in Appendix F.

Two primary categories of III-V semiconductor display and device production have attained economically feasible processing procedures—light-emitting diode (LED) displays and microwave IC devices.

LED's are fabricated from single-crystal GaAs in which p-n junctions are formed by the addition of suitable doping agents—typically silicon or chromium. Epitaxial layers of gallium arsenic phosphide (GaAsP) are grown on the substrate and result in an emission band of specific colors in the visible spectrum for displays and lower infrared for detectors. For example, red light with a peak at about 650 nm comes from the direct recombination of the p-n electrons and holes. Green emitting diodes are generally composed of gallium phosphide (GaP). The generalized LED processing steps (Table C1) are covered in the following section.

Microwave IC device processing is similar to silicon device manufacturing. Those process steps which warrant additional description are discussed in Appendix D.

III-V INGOT AND WAFER: LIGHT-EMITTING DIODES (LED)

Ingot Growing

As in the silicon ingot and wafer growth process, elemental forms of gallium and arsenic, plus small quantities of dopant material—silicon, tellurium, or chromium—are reacted at elevated temperatures to form ingots of doped single-crystal GaAs. Three generalized methods of ingot production are utilized:

1. Horizontal Bridgeman (HB)
2. Gradient Freeze (GF)
3. Liquid Encapsulated Czochralski (LEC Cz)

The bulk polycrystalline GaAs compound is normally formed by the reaction of As vapor with Ga metal at elevated temperatures in sealed quartz ampoules as shown in Fig. C1.

Typically, an As reservoir contained at one end of the ampoule is heated to approximately 600°C. This generates approximately 1 atm of As vapor pressure in the system, a prerequisite for obtaining stoichiometric GaAs. The As vapor reacts with the Ga metal which is maintained at approximately 1260°C and located at the other end of the ampoule in a quartz board. After the Ga has been completely reacted, single-crystal

growth may be initiated by programmed cooling (gradient freeze) or by physically moving either the ampoule or furnace to provide proper temperature gradients for growth (horizontal bridgeman). This indirect approach (As transport) for compounding and growth of GaAs is used because of the high vapor pressure of As at its melting point and at the melting point of GaAs, about 20 atm at 812°C, and 60 atm at 1238°C, respectively.

Table C1. III-V Ingot and Wafer Process: Light-Emitting Diodes (LED)

I.	INGOT GROWING
a.	Charge Preparation
b.	Ampoule Load and Seal
c.	Furnace Growth
d.	Ampoule Breakout
e.	Ingot and Remelt Sandblasting/Cleaning
II.	WAFER PROCESSING
a.	X-ray Diffraction
b.	Ingot Cropping
c.	Wafer Slicing
d.	Wafer Wash
e.	Lapping
f.	Polishing
III.	EPITAXIAL
a.	Reactor Load and Unload
b.	Reactor Cleaning
IV.	MOCVD
V.	DEVICE FABRICATION
a.	Nitride Deposition
b.	Photolithographic Process
c.	Wet Etching
d.	Plasma Etching and Ashing
e.	Diffusion
f.	Metallization
g.	Alloying
h.	Backlapping
i.	Final Test

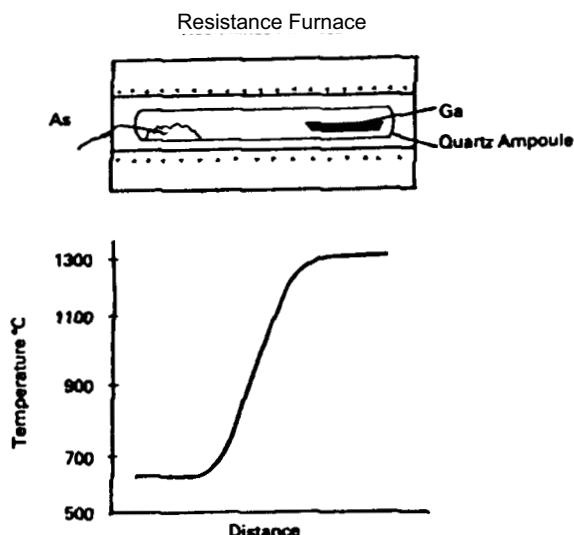


Figure C1. Sealed quartz ampoule GaAs growth system. (© 1981, CAL-OSHA, used with permission.)

Another approach to the commercial production of bulk single-crystal GaAs which is gaining favor is the liquid encapsulated Czochralski technique (LEC Cz), shown in Fig. C2. A Cz crystal puller is loaded with chunk GaAs in a quartz ampoule with an outer graphite receptor. The bulk GaAs is then melted at temperatures close to 1238°C, and the crystal is pulled in a pressurized atmosphere of approximately 100 atm. The melt is completely encapsulated by a viscous glass, B_2O_3 , which prevents melt dislocation when the dissociation vapor (As) is matched or exceeded by the pressure of an inert gas (argon, typically) applied in the puller chamber.

This description delineates processing steps in use in LED production facilities. The horizontal bridgman (HB) and gradient freeze (GF) methods of single-crystal gallium arsenide ingot growth are the dominant techniques in use in LED production.

Charge Preparation. A long cylindrical quartz ampoule is loaded with elemental arsenic (99.9999%) in chunk form in a locally-exhausted laminar flow hood. A smaller quartz boat is charged with frozen liquid gallium (99.9999%), elemental silicon dopant chunk polycrystalline GaAs, and a small single-crystal seed crystal surrounded with boron nitride powder. The quartz boat is then loaded into another section of the quartz ampoule already containing the elemental arsenic.

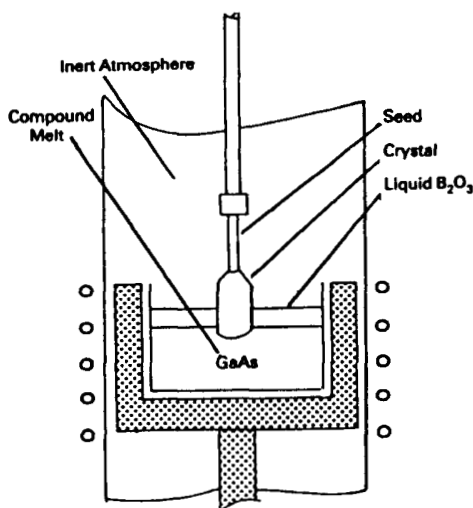


Figure C2. Liquid encapsulated Czochralski (LEC Cz) ingot growth system. (© 1981, CAL-OSHA, used with permission.)

Ampoule Load and Seal. The quartz ampoules are attached to an oil-diffusion vacuum pump assembly, evacuated, the ampoule heated with a hydrogen/oxygen torch, and finally, the end of the ampoule sealed. This creates a charged and sealed quartz ampoule ready for furnace growth.

Furnace Growth. The two methods of single-crystal ingot growth observed both utilize charged and sealed quartz ampoules in a high temperature furnace enclosure which is vented to a wet scrubber system.

The horizontal bridgman (HB) system consists of a two zone furnace in which the sealed quartz ampoule has separate temperature zones—the arsenic “cold” finger end at 610°C, and the quartz gallium/dopant/seed-crystal boat containing the melt at 1240°C. The basic principle in HB involves traversing two heated zones (one above the melting point of GaAs, and one below the melting point), over a boat of GaAs to provide the precisely controlled freezing of molten GaAs. The seed crystal, maintained at all times in the freeze zone, provides the initial crystal starting structure defining direction and orientation of the crystalline structure within the boat. The quartz boat and ampoule of gallium and arsenic are suspended within the heater chamber by a set of silicon-carbide liners called support tubes, which are positioned to allow the resistance heater assembly to mechanically move the full distance of the ampoule. Additionally, the ampoule rests on a table which must be raised and lowered during growth to provide the proper interface of the GaAs melt with the seed crystal.

Gradient freeze (GF) is the second ingot growth method in use. A zone high-temperature furnace utilizing resistance heating is kept at 1200–1300°C, with 1237°C being the melt/freeze point of GaAs. The total ingot growth process duration is three days, comprised of the following steps:

1. Furnace firing to temperature
2. GaAs synthesis
3. Seed melt
4. Cool down

The quartz ampoule is also raised and lowered during the growth process by the use of a scissor-type manual jack set-up.

Ampoule Breakout. After the single-crystal GaAs ingot is grown within the sealed quartz ampoule, the ampoule must be opened and the quartz boat containing the ingot plus seed crystal removed. This is accomplished by either cutting off the sealed end of the ampoule with a wet circular saw, or heating and cracking the ampoule with a hydrogen/oxygen torch. The quartz ampoules are recycled by wet etching the condensed arsenic on the interior surface with aqua regia (HCl/HNO_3), or sulfuric acid/hydrogen peroxide ($\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$).

Ingot and Remelt Sandblasting/Cleaning. The single-crystal GaAs ingots and polycrystalline defects must be sandblasted and cleaned to remove exterior oxides and contaminants. The sandblasting is done in a glove-box type bead blaster utilizing either silicon carbide or calcined alumina blasting media. Wet cleaning is done in wet chemical baths provided with local exhaust ventilation and utilizing aqua regia, or alcohol rinses (isopropyl alcohol and/or methanol).

Wafer Processing

X-Ray Diffraction. The crystalline orientation of the GaAs ingot must be determined by the use of an x-ray diffraction unit, as in silicon ingot processing.

Ingot Cropping. The ends or tails of the single-crystal ingot are removed using a water-lubricated single-bladed diamond saw, with various coolants added to the water.

Wafer Slicing. Gallium arsenide ingots are wax mounted to a graphite beam and sawed into individual wafers through the use of automatically operated inner diameter (ID) diamond blade saws. This operation

is done wet with the use of lubricants and generates a gallium arsenide slurry, which is collected, centrifuged, and recycled.

Wafer Wash. The non-cylindrically shaped GaAs wafers are dismounted from the wax and graphite beam. The wafers are then cleaned by sequentially dipping in wet chemical baths containing methanol, de-ionized water, and an acid mixture of sulfuric acid, hydrogen peroxide, and water (5:1:1). A visual inspection is also performed.

Lapping. Wafers are wax mounted to the lapper using a hot plate, and are lapped on a machine exerting a set rotational speed and pressure. A lapping solution is fed onto the lapping surface and constitutes a slurry of aluminum oxide, glycerin, and water. After a brief lapping period, the wafers are dismounted on a hot plate, rinsed in a soap solution and wiped dry.

Polishing. The wafers are then physically mounted to a mechanical polishing machine, using a sodium bicarbonate, 5% chlorine, water and colloidal silica slurry.

Epitaxial

The single-crystal GaAs wafers are used as substrates for the growth of very thin layers of the same or other III-V compounds having the desired electronic or optical properties. This must be done in such a way as to continue, in the grown layer, the crystal structure of the substrate. Such crystal growth, in which the substrate determines the crystallinity and orientation of the grown layer, is called epitaxy, and a variety of epitaxial growth techniques are used in III-V display and device production. The two most common techniques are:

1. Vapor phase epitaxy (VPE)
2. Liquid phase epitaxy (LPE)

VPE utilizes a heated stream of gaseous elements or compounds that interact at the surface of the substrate to form the crystalline layer and is primarily used in LED epitaxy.

In LPE, the same end is accomplished by cooling a heated metallic solution saturated with the components needed to grow the layer, while the solution is in contact with the substrate. This method is primarily used in microwave IC epitaxy.

In addition to VPE and LPE, vacuum epitaxy in the form of molecular beam epitaxy (MBE) has developed as an extraordinarily versatile technique. MBE of GaAs consists of an ultrahigh vacuum system containing sources for atomic or molecular beams of Ga and As and a heated

substrate wafer. The molecular beam sources are usually containers for liquid Ga or solid As. The sources have an orifice that faces the substrate wafer. When the container or effusion oven, as it is usually called, is heated, atoms of Ga or molecules of arsenic effuse from the orifice. For GaAs, growth usually takes place with a substrate temperature above 450°C.

Reactor Load and Unload. The degreased and polished wafers initially receive a *pre-epi* etch and clean step. This involves a sequential wet chemical dipping operation utilizing sulfuric acid, hydrogen peroxide, and water (5:1:1); a de-ionized water rinse; and finally, an isopropyl alcohol clean/dry.

Two major techniques of VPE are in use, based on two different chemistries:

1. The III-halogens (GaCl_3) and V-halogens (AsCl_3) or V-hydrogen (AsH_3 and PH_3)
2. The III metal-organics and V-hydrogen, such as $\text{Ga}_3(\text{CH}_3)$ and AsH_3

The thermochemistries of these techniques are very different. The halogen reactions (1) are usually “hot” to “cold” ones, in which the III-halogen is generated in a hot zone by reaction of the III element with HCl, and then diffuses to the cold zone, where it reacts with the V species to form III-V material. The metal-organic (2) chemistry is a *hot wall* process in which the III metal-organic compound *cracks* or pyrolyzes away the organic group and the remaining III and hydride V react to form III-V.

The primary technique in use for VPE in LED processing is the III-halogen and V-hydrogen system (1). It involves a two-cycle process—initially growing the epitaxial layer of GaAsP on the GaAs substrate, and lastly, an etch cycle to clean the quartz reactor chamber of impurities. During the epitaxial growth cycle, the pre-cleaned GaAs wafers are loaded into a vertical quartz reactor chamber containing an upper reservoir of elemental liquid gallium over which anhydrous HCl gas is metered, forming GaCl_3 . A V-hydrogen-hydride gas mixture of 10%- AsH_3 /5%- PH_3 / H_2 carrier is also metered into the reactor chamber with the addition of 50 ppm dimethyl telluride and 25 ppm diethyl telluride gaseous dopants. The chemical species in the “hot zone” of the quartz reactor react, and, in the “cold zone” form the desired layer of GaAsP on the wafer substrate as well as on the interior of the reactor chamber. Effluents from the reactor are

routed to a hydrogen torch system for pyrolysis and vented to a wet scrubber system or other exhaust conditioning system.

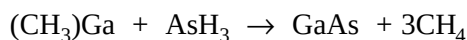
The etch cycle is performed at the end of the grow cycle and on new quartz reactors to clean the interior surface of impurities. Undiluted hydrogen chloride gas is metered into the chamber for periods of 5–15 minutes. The effluents are vented to the wet scrubber system for neutralization.

At the end of both the growth and etch cycles, an extended nitrogen purge is used to flush the reactor chamber of toxic and corrosive gases.

Reactor Cleaning. After each growth cycle, the VPE reactors must be opened, the wafers removed, and the lower portion of the reactor physically cleaned. The cleaning process is performed by the EPI operator. The lower quartz reactor and the bottom plate (base plate) are scraped clean using a metal tool, and the particulate material (mixture of GaAs, GaAsP, arsenic oxides, phosphorus oxides, and entrapped hydride gases) is collected in a metal container positioned below the vertical reactor. A high-efficiency vacuum is used for the final cleanup.

MOCVD

Metallorganic chemical vapor deposition (MOCVD) is another CVD option for III-V devices. While VPE refers to a compound material deposition process, MOCVD refers to the parent chemistry sources used in the system. Two chemistries are used, halides and metalorganic. The VPE process described above is a halide process. A group III halide (gallium) is formed in the hot zone and the III-V compound is deposited in the cold zone. In the metalorganic process for gallium arsenide, trimethylgallium is metered into the reaction chamber along with arsine, or a less toxic liquid alternative such as tertiarybutyl arsine, to form gallium arsenide. An example of a typical MOCVD reaction is:



Device Fabrication

The GaAs wafer with an epitaxially grown layer of GaAsP on the upper surface proceeds into the device fabrication processing sequence.

Nitride Deposition. A high-temperature chemical vapor deposition (CVD) of silicon nitride (Si_3N_4) is performed, using a standard diffusion

furnace. The gaseous sources are silane (SiH_4) and ammonia (NH_3) with a nitrogen carrier gas.

Photolithographic Process. The standard photoresist, aligning/exposure, developing and stripping process is utilized as in silicon device processing.

Wet Etching. Various mixtures of wet chemical acid solutions are used in plastic baths in locally-exhausted etch stations, some provided with laminar HEPA filtered supply systems vertically mounted. The primary acids in use are sulfuric (H_2SO_4), hydrofluoric (HF), hydrochloric (HCl) and phosphoric (H_3PO_4). As in silicon processing, hydrogen peroxide (H_2O_2) is used with sulfuric acid and ammonium hydroxide (NH_4OH) provides a caustic etch. A cyanide solution (sodium or potassium) is also used for etching aluminum.

Plasma Etching and Ashing. As an alternative to wet etching, plasma etching and ashing are used. The reactor configurations and reactant gases are very similar to those utilized in silicon device processing.

Diffusion. A closed ampoule zinc diarsenide solid source diffusion is performed in a vacuum diffusion furnace at 720°C , utilizing a nitrogen carrier gas.

Metallization. An initial aluminum evaporation is performed utilizing an e-beam evaporator. After backlapping, a last step, gold evaporation, is performed utilizing a filament evaporator.

Alloying. A final alloying step is performed in a low-temperature diffusion furnace, utilizing a nitrogen inert atmosphere.

Backlapping. Backlapping is done to remove deposited materials (GaAsP , Si_3N_4 , etc.) from the backside of the wafer. The wafers are wax-mounted to a lapper plate and wet-lapped with a colloidal silica slurry. Then the wax is removed by wet-stripping the wafers in an organic stripper containing phenol, sulfonic acid, and chlorobenzenes in a locally-exhausted wet chemical etch station. The wafers receive a final cleaning.

Final Test. As in silicon device processing, the completed LED circuits are computer-tested and marked. The final non-fabrication processing steps are generally performed at off-shore facilities, typically in the Far East.